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Chapter 1: Quantity and Unit

In this chapter you will:

- to identify physical quantities in daily life and then classify them into base and derived quantity
- to use the International Unit in measurement
- to simply convert the units of the length, mass, and time.
- to convert various units of base and derived quantity

We often hear expressions, such as the girl is beautiful, the scenery is pretty, the couple is in love, the singer's voice is melodious. In addition, we hear statements, including the height of the boy is 150 cm, the mass of an object is 50 kg, the speed of the car on the freeway is 80 km/hr, the journey from home to school requires 30 minutes. From several expressions and statements mentioned above, which ones are not measurable?

A. What is Physical Quantity?

Beautiful, pretty, love, and melodious cannot be measured nor stated in numbers, and therefore, they are not physical quantities.

Object observation must be based on an objective, honest and responsible.

NO	object	Measured (can/can't)	Measuring instrument
1	height	can	meter
2	hair color	can't	-
3	weight	can	scales
4	Body temperature	can	thermometer
5	Beauty/handsomeness	can't	-
6	smartness	can't	-
7	Speaking prowess	can't	-

On the other hand, height, mass, speed, and time can be measured and stated in numbers and therefore they are physical quantities.

Each physical quantity has a unit, for example: the height of that bridge is 8 meters.

height : physical quantity
8 : value
meter : unit

What is the meaning of measurement and how is it done? You have had experience in measuring objects, such as measuring the length of a line using a ruler. Other examples are: a carpenter measures the length of wood using a tape measure, and your physical education teacher counts your running time using a stopwatch.

Measurement is related to two important things, which are quantity and unit. In order to have a better understanding of quantity and unit, do the following activity.

Therefore, it can be concluded that:

Quantity is something that is measurable, generally has its own unit, and it is stated in numbers or value.

Unit is the standard of measurement.

Measuring is to compare the quantity measured with a similar quantity stated as units.

B. Base Quantity and Derived Quantity

Mass, length, and time are base quantities. The unit of mass, length, and time have been previously determined. The quantity of mass, length, and time are not derived from other quantities and therefore they are called *base quantities*. Thus, the base quantity is the quantity whose unit has been previously determined and it is not derived from another quantity.

Based on the International System of Units, there are seven base quantities. These are:

Besaran Pokok	satuan internasional (SI)	Alat ukur
Length	meter (m)	Ruler, vernier caliper, micrometer caliper
Mass	kilogram (kg)	Balance scale
Time	Kelvin (K)	Stopwatch, watch
Temperature	detik (s)	Thermometer
Electric current	Ampere (A)	Amperemeter
Luminous intensity	candela (cd)	Photometer
Amount of substance	mole (mol)	

Quantities such as height, width, radius, circumference, and distance are length quantities although they have different terms. Other quantities such as area, volume, and speed are *derived quantities*, since their units are derived from a base quantity unit. Therefore, a derived quantity is a quantity whose unit is derived from a base quantity.

Example:

1. Area is derived from length.

$$\begin{aligned} \text{Area} &= \text{length} \times \text{width} \\ &= \text{side} \times \text{side} \text{ (for a square)} \\ &= m \times m \\ &= m^2 \end{aligned}$$

2. Volume is derived from length.

$$\begin{aligned} \text{Volume} &= \text{length} \times \text{width} \times \text{height} \text{ (for a block)} \\ &= \text{side} \times \text{side} \times \text{side} \text{ (for a block)} \\ &= m \times m \times m = m^3 \end{aligned}$$

3. Velocity is derived from length and time.

$$Velocity = \frac{distance}{time}$$

$$Velocity = \frac{length}{time} = \frac{m}{s} \text{ or } \frac{m}{s}$$

Quantities are considered as derived quantities:

Quantity	Equation	Unit	Unit symbol	
Area	Area = length $\times$ width	square metre		m ²
Volume	Volume = length $\times$ width $\times$ height	cubic metre		m ³
Frequency	$frequency = \frac{1}{period}$	hertz	Hz	s ⁻¹
Density	$density = \frac{mass}{volume}$	kilogram per cubic metre		kg m ⁻³
Velocity/speed	$velocity = \frac{displacement}{time}$	metre per second		m s ⁻¹
Acceleration	$acceleration = \frac{velocity\ change}{time}$	meters per second squared		m s ⁻²
Force	Force = mass $\times$ acceleration	Newton	N	$\frac{m}{s^2} \text{ kg}$
Pressure, stress	$pressure = \frac{force}{area}$	Pascal	Pa	N m ⁻²
Energy, work	Work = force $\times$ displacement	Joule	J	N m
Power	$power = \frac{work}{time}$	Watt	W	J s ⁻¹
Quantity of electricity	Quantity of charge = current $\times$ time	Coulomb	C	A s
Potential difference, electromotive force	$potential\ difference = \frac{energy}{charge}$	Volt	V	W A ⁻¹
Electric resistance	$resistance = \frac{potential\ difference}{current}$	Ohm	Ω	V A ⁻¹

C. International System of Units

In France, in the year 1866, the use of a system of measurement was ratified. Later it was adopted by other countries. In 1960, the *International System of Units* abbreviated SI. The objectives were to create agreements between nations through equalizing and facilitating the use of measuring instruments, thus making conversion easier all over the world.

Why do we use SI? SI is used to obtain some facilities as follows:

1. In SI, every quantity has only one base unit. Bigger or smaller units are associated with the base unit by adding prefixes.
2. SI units are based on the Decimal System, which uses multiples of 10.

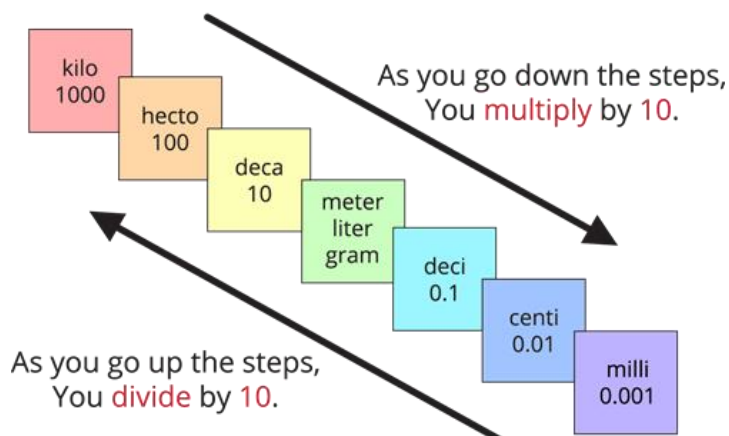
Look at the table below presenting SI units based on the decimal system

Prefixes Used with SI Units				
Prefix	Symbol	Factor	Scientific notation	Example
giga	G	1 000 000 000	10^9	gigameter (Gm)
mega	M	1 000 000	10^6	megagram (Mg)
kilo	k	1000	10^3	kilometer (km)
deci	d	1/10	10^{-1}	deciliter (dL)
centi	c	1/100	10^{-2}	centimeter (cm)
milli	m	1/1000	10^{-3}	milligram (mg)
micro	μ	1/1 000 000	10^{-6}	microgram (μ g)
nano	n	1/1 000 000 000	10^{-9}	nanometer (nm)
pico	p	1/1 000 000 000 000	10^{-12}	picometer (pm)

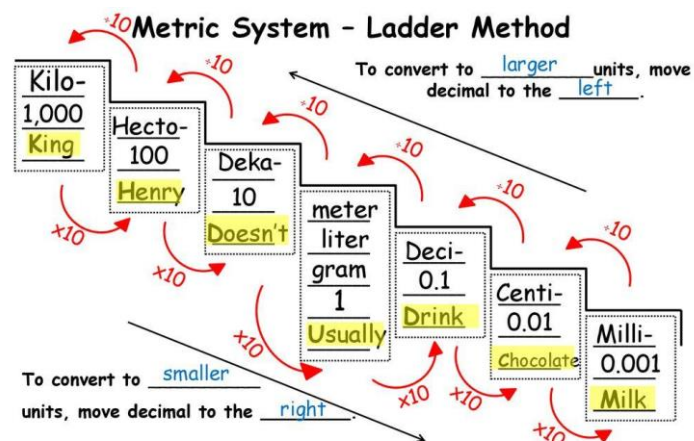
D. Conversion of Length, Mass, and Time Units

In order to state a distance between two cities, we use the kilometer (km) unit. Meanwhile, if the distance between two places is not too far, the meter (m) unit can be used. For example, the distance between the football field and school is 0,5 km. Converting a unit into another will produce conversion.

What method do we use to convert length and mass units in SI? The simplest way is by using a conversion ladder where every step down is multiplied by 10 and every step up is divided by 10. Look at the picture below:



The conversion of units in SI using the conversion ladder is as follow:



1. Conversion of a number down the conversion ladder

- a. For one step down the conversion ladder, the initial number is multiplied by 10.

Example:

The initial unit is expressed as a meter (m) and you want to convert it into a decimeter (dm). You must multiply the initial number by 10, since we are taking 1 step down from m to dm. Thus, $1 \text{ m} = 1 \times (10) \text{ dm} = 10 \text{ dm}$.

- b. For two steps down the conversion ladder, the initial number must be multiplied by 100.

Example:

The initial unit is in meters (m) and you want to convert it into centimeters (cm). You must multiply the initial number by 100, since it takes 2 steps down from m to cm. Thus, $4.5 \text{ m} = 4.5 \times (100) \text{ cm} = 450 \text{ cm}$.

2. Conversion of a number up the conversion ladder

- a. For one step up the conversion ladder, the initial number must be divided by 10.

Example:

The initial unit is a decagram (dag) and you want to convert it into a hectogram (hg). You must divide the initial number by 10, since it takes 1 step upward from dag to hg. Thus, $1 \text{ dag} = 1 : (10) \text{ hg} = 0.1 \text{ hg}$.

- b. For two steps up the conversion ladder, the initial number must be divided by 100.

Example:

The initial unit is a gram (g) and you want to convert it into a hectogram (hg). You must divide the initial number by 100, since it takes 2 steps to go up from g to hg. Thus, $500 \text{ g} = 500 : (100) \text{ hg} = 5 \text{ hg}$.

3. Time Unit Conversion

In the time unit, conversion cannot be conducted using a conversion ladder. The time units which have been internationally acknowledged are as follows.

1 day = 24 hours

1 hour = 60 minutes

1 minute = 60 seconds

Self Assessment

Fill in the blanks.

$$1 \text{ dag} = \dots \text{ mg}$$

$$\dots \text{ cg} = 0.5 \text{ dag}$$

$$5 \text{ g} = \dots \text{ kg}$$

$$\dots \text{ dL} = 800 \text{ mL}$$

$$\dots \text{ daL} = 60 \text{ L}$$

$$1 \text{ m} = \dots \mu\text{m}$$

$$\dots \text{ mm} = 10 \mu\text{m}$$

$$\dots \text{ second(s)} = 2 \text{ minutes}$$

$$\dots \text{ hour(s)} = 150 \text{ seconds}$$

$$\dots \text{ minute(s)} = 1.5 \text{ hours}$$

E. Conversion of Derived Quantity

Similar to the base quantity, the derived quantity can also be converted. Look at the example below.

1. Area Unit

In order to determine an area of a rectangle, the measurement is conducted with an indirect method as follows:

$$\text{Area of a rectangle} = \text{length} \times \text{width}$$

$$= m \times m = m^2$$

- a. If the unit is converted to cm^2 , the method is as follows:

$$1 \text{ m}^2 = 1 \text{ m} \times 1 \text{ m}$$

$$= 100 \text{ cm} \times 100 \text{ cm}$$

$$= 10,000 \text{ cm}^2$$

- b. If the unit is converted to dam^2 , the method is as follows:

$$1 \text{ m}^2 = 1 \text{ m} \times 1 \text{ m}$$

$$= 0.1 \text{ dam} \times 0.1 \text{ dam}$$

$$= 0.01 \text{ dam}^2$$

Now, try to convert the unit to dm^2 and km^2 .

2. Volume Unit

In order to determine the volume of a block, the measurement is conducted with an indirect method as follow:

$$\text{Volume of a block} = \text{length} \times \text{width} \times \text{height}$$

$$= m \times m \times m = m^3$$

- a. If the unit is converted to cm^3 , the method is as follows:

$$1 \text{ m}^3 = 1 \text{ m} \times 1 \text{ m} \times 1 \text{ m}$$

$$= 100 \text{ cm} \times 100 \text{ cm} \times 100 \text{ cm}$$

$$= 1,000,000 \text{ cm}^3$$

b. If the unit is converted to dam^3 , the method is as follows:

$$\begin{aligned}1\text{ m}^3 &= 1\text{ m} \times 1\text{ m} \times 1\text{ m} \\&= 0.1\text{ dam} \times 0.1\text{ dam} \times 0.1\text{ dam} \\&= 0.001\text{ dam}^3\end{aligned}$$

Now, try to convert the unit to dm^3 and km^3 .

3. Velocity Unit

Velocity can be measured directly using the speedometer attached to vehicles. In order to know the velocity of a vehicle, you just need to look at the speedometer. The measurement of velocity can be indirectly conducted by measuring distance (length) and travel time. For example: the distance (length) is measured in km and the time is measured in hours. The formula can be written as follows.

$$\text{Velocity} = \frac{\text{distance}}{\text{time}} = \frac{\text{km}}{\text{hour}} = \frac{\text{km}}{\text{hour}}$$

If km/hour unit is converted to m/s (meter/second), the method is as follows.

$$1 \frac{\text{km}}{\text{hour}} = \frac{1\text{ km}}{1\text{ hour}} = \frac{1,000\text{ m}}{3,600\text{ s}} = 0.2777 \frac{\text{m}}{\text{s}}$$

$$\text{Thus, } 1 \frac{\text{m}}{\text{s}} = \frac{1}{0.27777} \times 1 \frac{\text{km}}{\text{hour}} = 3.6 \frac{\text{km}}{\text{hour}}$$

F. Expressing Physical Unit

There are some units in physics taken from inventor's names, for example the unit of power, a watt, is taken from the name of the inventor of the steam engine, **James Watt**. However, there are also other units whose names are not taken from the names of inventors, for example kilogram, meter, and second. In order to differentiate between the name of a unit and the name of an inventor, there are rules in writing units.

Basically, units can be written as abbreviations (symbols) or fully written with lowercase letters. When a unit's name is derived from its inventor's name, if it is fully written, it must be written in lowercase. However, if we use a symbol, we must use a capital.

Look at these examples.

1. Units which are not derived from inventor's names.

- Mass : kilogram, symbol: kg
: gram, symbol: g
- Length : meter, symbol: m
: kilometer, symbol: km
: hectometer, symbol: hm
: decameter, symbol: dam
: centimeter, symbol: cm
: millimeter, symbol: mm
- Time : second, symbol: s
- Velocity : meter per second, symbol: km/hour
- Acceleration : meter per second squared, symbol: m/s^2

- Density kilogram per cubic meter, symbol: kg/m^3

2. Units which are derived from inventor's names.

- Electric current : ampere, symbol: A
- Electric current : milliampere, symbol: mA
- Force : newton, symbol: N
- Power : watt, symbol: W
- Work : joule, symbol: J
- Electrical Energy : kilowatt-hour, symbol: kWh

3. STANDARDS OF LENGTH, MASS, AND TIME

3.1. Measurement of length

To measure the quantity of length, we can use a ruler, vernier caliper or micrometer screw gauge. However, those instruments mentioned have their own strengths and weaknesses

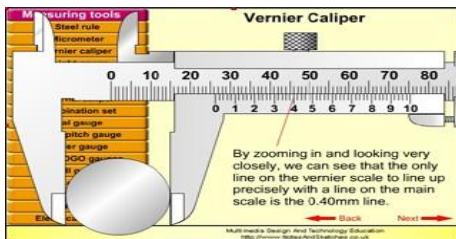
a. Ruler

Ruler is an instrument to measure length which usually has ***smalest scale of 0,1 cm.***



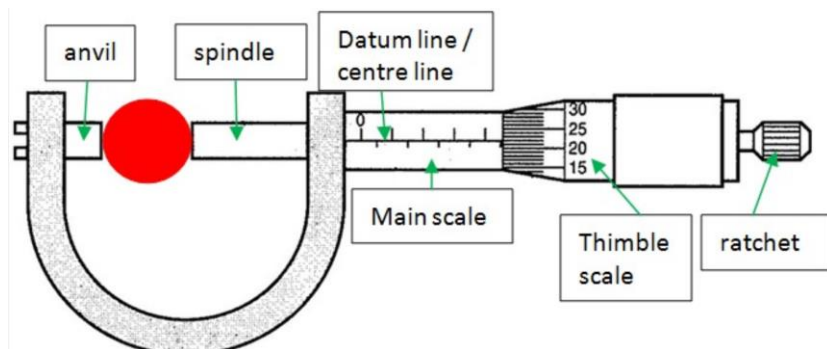
b. Vernier caliper

Vernier caliper is an instrument to measure length which is more accurate than a ruler since it has the smallest scale of 0,1 mm or **0,01 cm**, used for measure diameter of an object (circle/cylinder)



c. Micrometer screw gauge

A micrometer screw gauge is an instrument to measure length which is more accurate than a ruler or a vernier caliper, because it has the smallest scale of **0,01 mm**, used for thickness.



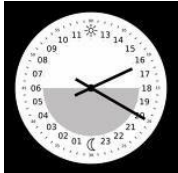
3.2. Measurement of Mass

To measure the quantity of mass, we can use balances. There are several balances that are often used, such as cross-beam balance, spring balance, O – hauss balance and digital balances



3.3. Measurement of Time

To measure the quantity of time, we can use the instruments like wristwatches, clocks and stopwatches



3.4. Measuring Cup

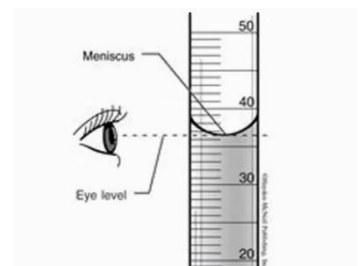
The scale on the measuring cup is actually when you see the liquid in the measuring cup, it will look concave. So this concave look is called a concave meniscus.

What is the Meniscus? The meniscus is the curvature of the surface of a liquid in the tube. Some forms of curvature are convex and some are concave.

The concave meniscus occurs because the attractive force between the liquid and glass particles (adhesion) is greater than the attractive force between the liquid particles (cohesion).

There is something we need to pay attention to when we read the scale on a measuring cup, namely the eye must be straight in the liquid basin. Look at the picture below:

If our eyes see the curvature of the liquid from above or below it will show different results. So it must be remembered that the eyes must be straight in the hollow of the liquid.



Practice quantities, units and measurements

I. Unit conversion

Remember the math formula

$$10^a \times 10^b = 10^{a+b} \quad \text{example: } 2000 \text{ hm} = \dots \text{ mm} \rightarrow 2 \times 10^3 \times 10^5 = 2 \times 10^{3+5} = 2 \times 10^8 \text{ mm}$$

$$10^c : 10^d = 10^{c-d} \quad \text{example : } 300,000 \text{ mm} = \dots \text{ Mm} \rightarrow 3 \times 10^5 : 10^9 = 3 \times 10^{5-9} = 3 \times 10^{-4} \text{ Mm}$$

$$\text{Or } 3 \times 10^5 \times 10^{-9} = 3 \times 10^{-4} \text{ Mm}$$

Exercise

1. $7 \text{ km} = \text{m} = \dots \mu\text{m} = \dots \text{Gm}$
2. $3 \times 10^6 \text{ g} = \dots \text{mg} = \dots \text{tons}$
3. $500 \text{ seconds} = \dots \text{minutes} = \dots \text{hours}$
4. $48 \text{ hours} = \dots \text{seconds} = \dots \text{days}$
5. $4 \text{ dam}^2 = \dots \text{cm}^2 = \dots \text{km}^2 = \dots \text{dm}^2$

6. $9,000 \text{ cm}^3 = \dots \text{ L} = \dots \text{ m}^3$

7. a. $20 \text{ g/cm}^3 = \dots \text{ kg/m}^3$

b. $5 \text{ g/cm}^3 = \dots \text{ kg/m}^3$

8. a. $5000 \text{ kg/m}^3 = \dots \text{ g/cm}^3$

b. $7.5 \times 10^6 \text{ kg/m}^3 = \dots \text{ g/cm}^3$

9. $72 \text{ km/hour} = \dots \text{ m/s}$

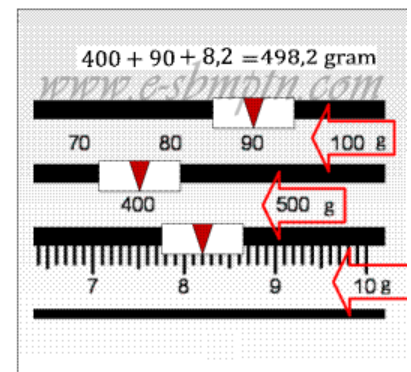
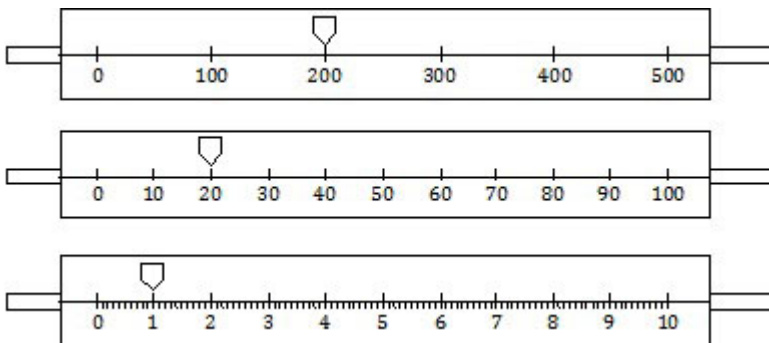
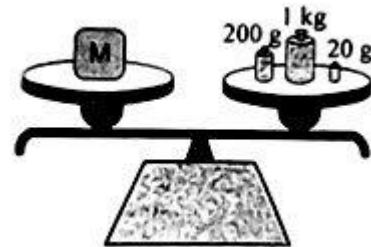
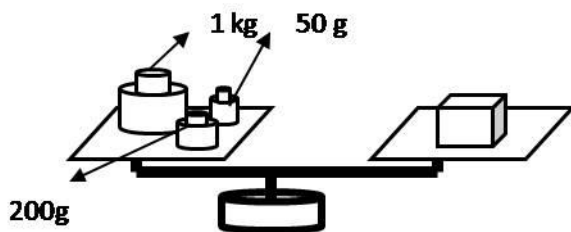
b. $18 \text{ km/h} = \dots \text{ m/s}$

$103 \text{ m/s} = \dots \text{ km/hour}$

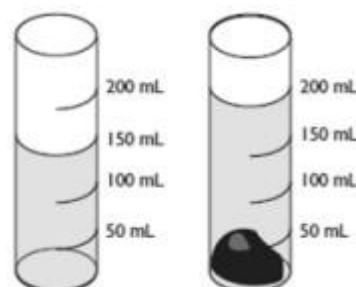
b. $5 \text{ m/s} = \dots \text{ km/h}$

II. Measurement

1. Balance / scales



2. measuring cup



III. HOTS

- The inner side of a bathtub has a length of 120 cm, a width of 100 cm and a height of 100 cm.
 - What is the volume of the bath if only $\frac{3}{4}$ of the volume of the bath is filled in liters?
 - The tub is drained using a bucket with a volume of 5 dm^3 until it runs out. How many times should the bucket be used?
- The speed of light is 300 million m/s. How many times faster is the speed of light when compared to the speed of a Formula 1 car, which is 250 km/hour?

WORKSHEET PHYSICS

Density

Aim of Practice

- Participant educate can determine big mass type object
- Participant educate can use o- hauss balance For measure mass object
- Participant educate can use glass measuring For measure the volume of objects No order
- Participant educate can use ruler For measure the volume of objects order

2. Tools and Materials

- O- hauss balance 3 arms
- Measuring cup
- Ruler
- Cube
- Rock

3. How it works

- Measure long side cube with use ruler
- Measure mass cube with use o- hauss balance
- noted results measurement mass and length side cube
- Measure mass rock with use O-haus balance
- Measure the volume of the rock with use glass measuring
- noted results measurement rock mass and volume
- Mengh that's it mass type cubes and stones

4. Results of practice and calculations

4.1. Cube

4.1.1. The mass of a cube = 256 grams = $256 : 1000 = 0.256 \text{ kg}$

4.1.2. side length cube = 2 cm = 0.02 m

Volume of cube = side x side x side = $2 \times 2 \times 2 = 8 \text{ cm}^3 =$
 $8 : 1000,000 = 0.000 008 \text{ m}^3 = 8 \times 10^{-6} \text{ m}^3$

4.1.3. Density = mass : volume =

$$256 : 8 = 32 \text{ g/cm}^3 = 32 \times 1000 = 32000 \text{ kg/m}^3$$

4.2. Rock

4.2.1. Stone mass = grams = kg

4.2.2. Initial volume of water =

Final volume of water =

Rock volume = Final volume – initial volume =

4.2.3. Density = mass : volume =

5. Conclusion

5. 1. Mass type cube is kg/m^3 . Based on table mass type obtained results that cube the is

5.2. Density _ rock is kg/m^3 . Based on table mass type obtained results that rock the is

Material	Density (g/cm ³)	Material	Density (g/cm ³)
Aluminum	2.64	Iron (cast)	7.21
Brass	8.55	Iron (wrought)	7.77
Brick (red, common)	1.92	Lead	11.34
Coal (anthracite)	1.51	Marble	2.56
Concrete	2.37	Paraffin (wax)	0.72
Copper (cast)	8.68	Quartz	2.64
Copper (rolled)	8.91	Rubber	1.52
Cork	0.24	Steel (cast)	7.85
Feldspar	2.56	Steel (rolled)	7.93
Glass (window)	2.58	Wood (dry) - red cedar	0.38
Gneiss	2.87	Wood (dry) - Douglas fir	0.53
Granite	2.69	Wood (dry) - hickory	0.85
Gold (pure, 24 kt)	19.29	Wood (dry) - maple	0.70
Ice	0.92	Wood (dry) - red oak	0.70
Ivory	1.84	Wood (dry) - yellow pine	0.70

Chapter 2: Temperature

In this chapter you will:

- to be able to give reasons why sensory receptors cannot be used to measure temperature.
- To use a thermometer to measure the temperature of any substance.
- To be able to make a simple-scaled thermometer based on the characteristics of volume change when a liquid receives heat.
- To be able to compare a Celcius thermometer scale with other thermometer scales.

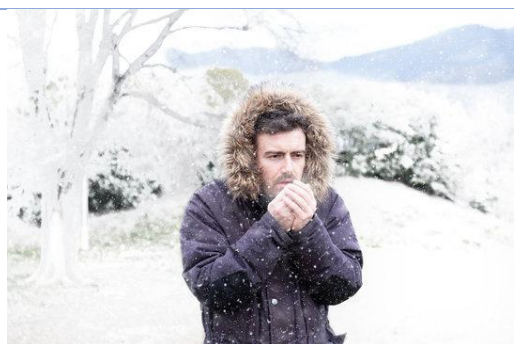


In the desert, it is very hot during the day but very cold during the night.

When you are on a mountain, you might feel cold. When you are on a beach, you might feel warm. What do cold and warm mean? Can you measure the temperature of hot and cold using your sensory receptors? What instrument is needed to measure temperature accurately?

A. What is Temperature?

People can feel either (a) cold or (b) warm.



(a) cold



(b) warm

What do you feel when you drink icy water? It is cold, isn't it? When you are boiling water, eventually the water heats up, won't it? How do you differentiate between hot and cold?

In order to know for sure how hot or how cold an object is, you need a quantity that is measurable by instruments. The quantity related to the hotness or coldness of an object is called *temperature*. Therefore, temperature is a quantity stating how hot or cold an object is.

You already know that the quantity stating the degree of heat and cold of an object is called temperature. Now, the question is: how do you measure temperature? Can you use your hand to measure hot and cold accurately?

Self Assessment

Why do hands, which have been dipped in warm water and then dipped in plain water, feel cool?

Why do hands, which have been dipped in cold water and then dipped in plain water, feel warm?

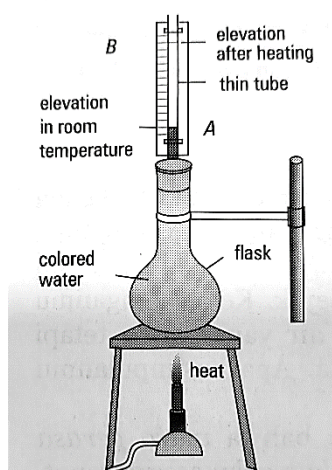
Why can't you use your hands to measure temperature accurately?

B. Thermometer

The sensory receptors in your hands cannot be used to measure temperature accurately. Therefore, you must use a *thermometer*.

To accurately measure a temperature, a thermometer must have particular characteristics. Those characteristics are:

1. The scale is easy to read.
2. It is sensitive to temperature change.
3. The coverage is sufficiently wide.
4. It is safe to use.



The principle used in a thermometer is the change of liquid volume. If a flask is filled with colored water until height A in the thin tube and then heated, the flask and the water will expand. Water expansion is higher than the flask's expansion and therefore the water in the tube will rise up to height B. This shows that the change of liquid (water) volume in the thin tube can be used to measure temperature.

C. Making Thermometer Scale

Thermometer can be used as a measuring instrument if it has a scale. How do we make a scale on a thermometer?

The method to make a scale on a thermometer is called *calibration*. The methods are as follows.

1. Deciding the lower fixed point

Put the unscaled thermometer's sensing portion in a container filled with melting ice. Let it sit there for a moment. Watch the mercury in the thermometer carefully. If the mercury's surface is stable, then it means that the mercury's temperature is equal to the temperature of melting ice water. Mark the fixed point by writing 0°C .

2. Deciding the upper fixed point

Put the mercury thermometer into boiling water. Let it there for a moment and watch the mercury's surface. If the mercury's surface is stable, it means that the mercury's temperature is equal to the temperature of boiling water. Mark the fixed point by writing 100°C .

3. Dividing the interval between lower fixed point and upper fixed point into 100 parts

Divide the interval between the lower fixed point (0°C) and the upper fixed point (100°C) into 100 parts and mark them. Each part represents 1°C .

4. Extended thermometer scale

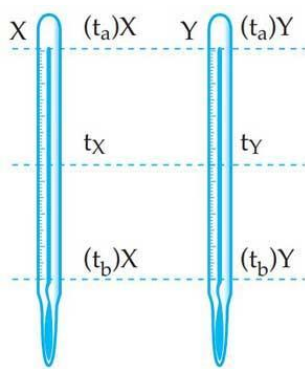
The thermometer scale can be extended below 0°C until it reaches -10°C and above 100°C up to 110°C.

In order to have a better understanding concerning how to make a scale on a thermometer.

Non-standard thermometer scale conversion

D. Thermometer scale conversion

In general, the relationship between the thermometer scales with one another is as shown in the following figure.



Thermometer x with lower fixed point (tb)X and upper fixed point (ta)X. Thermometer Y with a lower fixed point (tb)Y and an upper fixed point (ta)Y. The lower fixed point and upper fixed point of the two thermometers above are the temperature when ice melts and the temperature when water boils at 1 atmosphere pressure.

By comparing the temperature change and the interval between the two fixed points of each thermometer, the following relationship is obtained.

$$\frac{(t_a)X - t_X}{(t_a)X - (t_b)X} = \frac{(t_a)Y - t_Y}{(t_a)Y - (t_b)Y}$$

Information:

- (ta)X = boiling point on thermometer X
- (tb)X = freezing point of thermometer X
- tX = temperature on thermometer X
- (ta)Y = boiling point on thermometer Y
- (tb)Y = freezing point of thermometer Y
- tY = temperature on thermometer Y

Example:

- Thermometer X is designed to measure frozen water on a scale of -40 and boiling water on a scale of 160. If an object is measured with a Reamur thermometer, it shows a value of 20°R, then determine the designated value when measured with thermometer X!

Solution:

$$(t_b)X = -40^{\circ}X$$

$$(t_a)X = 160^{\circ}X$$

$$t_R = 20$$

The boiling point Reamur thermometer is 80°R and freezing point is 0°R so:

$$(t_a)R = 0^{\circ}R$$

$$(t_b)R = 80^{\circ}R$$

$$t_X \dots ?$$

Answer:

$$\frac{(t_a)X - t_X}{(t_a)X - (t_b)X} = \frac{(t_a)R - t_R}{(t_a)R - (t_b)R}$$

$$\frac{160 - t_X}{160 - (-40)} = \frac{80 - 20}{80 - 0}$$

$$\frac{160 - t_X}{200} = \frac{60}{80}$$

$$\frac{160 - t_x}{200} = \frac{6}{8}$$

$$8(160 - t_x) = (200)(6)$$

$$1280 - 8t_x = 1200$$

$$8t_x = 1280 - 1200$$

$$8t_x = 80$$

$$t_x = 80/8$$

$$t_x = 10$$

So, the temperature of the object when measured with a thermometer X is 10°X .

2. Two thermometers X and Y have ice points marked at 15° and 25° and steam points marked as 75° and 125° respectively. When thermometer X measures the temperature of a bath as 60° on it, what would thermometer Y read when it is used to measure the temperature of the same bath?

Solution:

The correct option is **A** 100°



$$\frac{X - 15^\circ}{75^\circ - 15^\circ} = \frac{Y - 25^\circ}{125^\circ - 25^\circ}$$

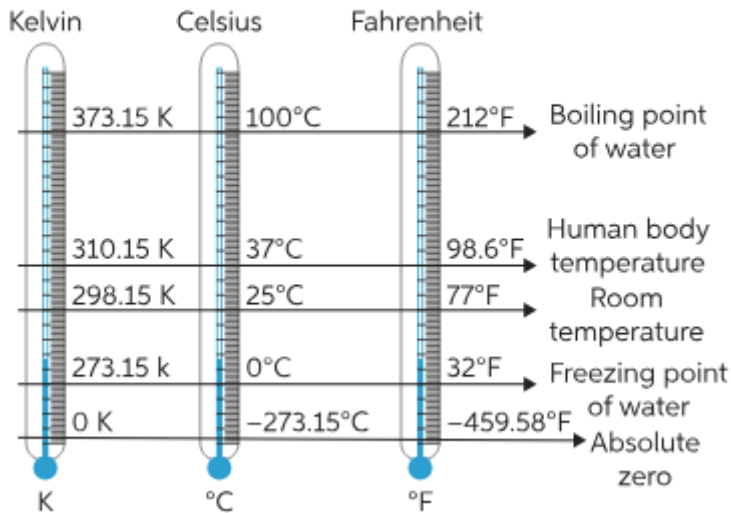
Given, $X = 60^\circ$

$$\frac{60^\circ - 15^\circ}{75^\circ - 15^\circ} = \frac{Y - 25^\circ}{125^\circ - 25^\circ}$$

$$\frac{45^\circ}{60^\circ} = \frac{Y - 25^\circ}{100^\circ}$$

$$Y = 100^\circ$$

Comparison of three different temperature scales



Temperature Scales

The thermometer calibration process described above results in what is known as a centigrade thermometer. A centigrade thermometer has 100 divisions or intervals between the normal freezing point and the normal boiling point of water.

The conversion scale in thermometer Celcius, Reamur, Fahrenheit, Kelvin

From	To			
	Celcius	Reamur	Fahrenheit	Kelvin
Celsius		$\frac{4}{5}C$	$\frac{9}{5}C + 32$	$C + 273$
Reamur	$\frac{5}{4}R$		$\frac{9}{4}R + 32$	$\frac{5}{4}R + 273$
Fahrenheit	$\frac{5}{9}(F - 32)$	$\frac{4}{9}(F - 32)$		$\frac{5}{9}(F - 32) + 273$
Kelvin	$K - 273$	$\frac{4}{5}(K - 273)$	$\frac{9}{5}(K - 273) + 32$	

Example:

1. The temperature of object is 122°F. What is the temperature in degrees Celsius?

Anwer:

$$T_C = \frac{5}{9}(F - 32)$$

$$T_C = \frac{5}{9}(122 - 32) = T_C = \frac{5}{9}(90) = 5(10)$$

$$T_C = 50^\circ\text{C}$$

$$\begin{aligned}
 2. \quad 212^{\circ}\text{F} &= \dots \text{K} \\
 T^{\circ}\text{C} &= (100/180)(T^{\circ}\text{F} - 32) \\
 T^{\circ}\text{C} &= (5/9)(T^{\circ}\text{F} - 32) \\
 T^{\circ}\text{C} &= (5/9)(212 - 32) \\
 T^{\circ}\text{C} &= (5/9)(180) \\
 T^{\circ}\text{C} &= (5)(20) \\
 T^{\circ}\text{C} &= 100
 \end{aligned}$$

E. Various Thermometers in Daily Life

Types of Thermometer

Following are the different types of thermometers that we use on a daily basis:

- Clinical thermometer /medical thermometer
- Laboratory thermometer
- Digital thermometer
- Infrared ear thermometer
- Mercury thermometer
- Alcohol in glass thermometer
- Constant pressure gas thermometer
- Constant volume thermometer
- Platinum resistance thermometer
- Thermocouple thermometer
- Pyrometer thermometer

Clinical Thermometer

CLINICAL THERMOMETER

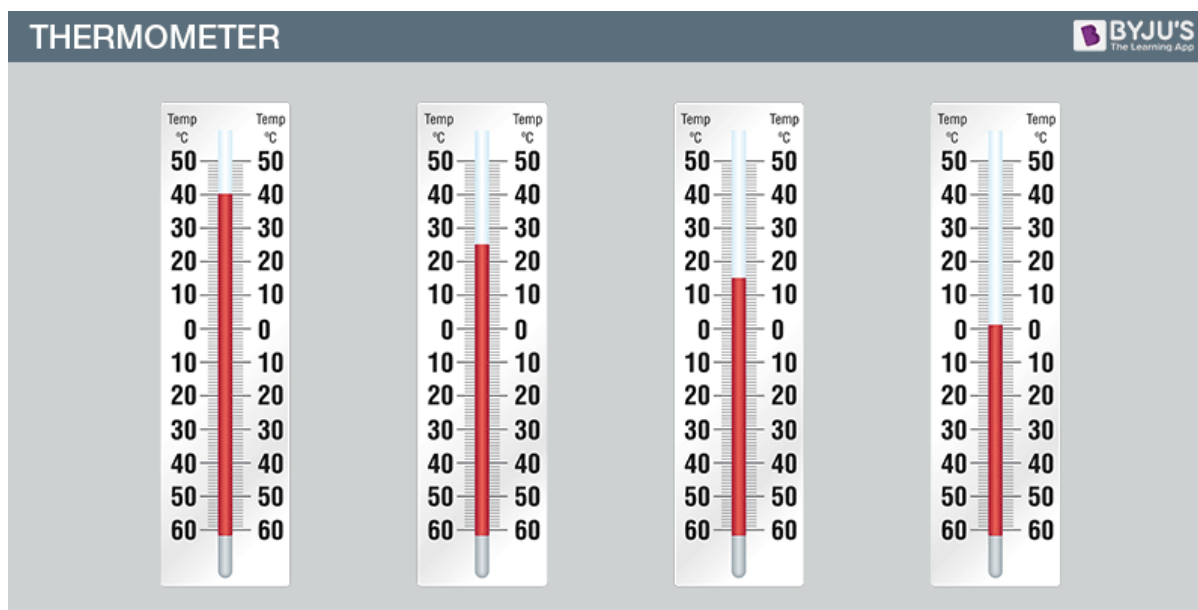


- Clinical thermometers are meant for clinical purposes.
- It is developed for measuring the human body temperature.
- It is a long narrow glass tube with a bulb containing mercury at the end.
- The normal human body temperature is 37°C , which can fluctuate between the ranges 35°C to 42°C . Hence, the clinical thermometers range from 35°C to 42°C .
- The level of mercury tells our body temperature in $^{\circ}\text{C}$.
- Since mercury is a toxic element, thus these thermometers have been replaced by digital thermometers nowadays.

Note: Clinical thermometers must be sterilized before use for a safe and clean check-up.

How to Read a Thermometer?

- The thermometer is washed with water at a normal temperature.
- It is given a few jerks. We notice that the jerks bring the level of mercury down. We ensure that it falls below the 37°C mark or the 98°F mark, the normal human body temperature.
- The thermometer is then kept under the tongue or the arms to get the reading on the thermometer.
- The thermometer is then held close to the eye, as shown in figure 3, to see the correct reading.
- The temperature difference indicated between the two bigger marks is noted. Also, the number of divisions (shown by smaller marks) between these marks is noted. For e.g. the bigger mark reads 10 and the number of divisions is 5, then each division reads the value equal to $10/5 = 20$



- We note the value of the bigger mark that the mercury thread crossed (98 in Fig.4 Example 1). Then the number of the division of the mercury thread has crossed is noted (3 in this case). Now, this value is multiplied by the value of one division (which is 0.2 in this case). Hence, the temperature reading is $98 + (3 \times 0.2) = 98.6$.

Laboratory Thermometer

- Since clinical thermometers can't be used to measure temperature other than the human body, we need a special type of thermometer for other purposes.
- A laboratory thermometer, which is colloquially known as the lab thermometer, is used for measuring temperatures other than the human body temperature.
- It ranges from -10°C to 110°C.
- Laboratory thermometer Uses:
 - Designed for lab purposes
 - Checking the boiling point
 - Checking freezing point
 - Checking the temperature of other substances.

- Checking the temperature of a solvent but not a clinical one.

LABORATORY THERMOMETER



Weather reporters also use maximum-minimum thermometers for measuring maximum and minimum temperatures of a place.

Learn more about thermometers, [heat energy](#), and temperature with BYJU'S engaging video lectures.

Digital Thermometer

- These thermometers measure the temperature through an electronic circuit.
- The information captured is sent to a microchip that processes it and displayed numerically on the digital screen.
- They are easy to use, inexpensive, and accurate.
- Digital thermometers can be considered advanced thermometers used for measuring body temperature.

How to Use a Digital Thermometer?

- Turn on the thermometer and click till the number zero appears on the display.
- Place the tip of the thermometer under the armpit.
- Wait for a few seconds before you hear the alarm from the thermometer.
- Remove the thermometer and check for the temperature on the display.
- After use, clean the tip of the thermometer with cotton and alcohol.

Infrared Ear Thermometer

- Checking the body temperature using the inside of the ear is known as **tympanic membrane temperature**.
- This method is common when the temperature needs to be taken from the children as it is quick.
- An infrared thermometer captures the temperature in the form of infrared energy given off by the heat source.

How to Use an Infrared Ear Thermometer?

- The tip of the thermometer should be placed inside the ear, pointing toward the nose.
- The power button on the thermometer should be pressed until a signal is heard.
- Read the temperature value.
- Remove the thermometer from the ear and clean the tip with cotton and alcohol.
- Make sure the thermometer is not used for the ear containing a lot of wax.
- Thermometers such as mercury thermometers, alcohol in glass thermometers, constant pressure gas thermometers, and platinum resistance thermometers are designed based on technology.

Mercury Thermometer

- These thermometers are not used in some sectors since they can easily break.
- Mercury is a poisonous liquid that can cause health hazards.
- Mercury thermometers are used to provide accurate temperature readings.
- It is used to measure the temperature under the arm, orally and rectally.

Alcohol In Glass Thermometer

- These are thermometers composed of sealed glass.
- Alcohol In Glass Thermometer shows the temperature by the level at which mercury or alcohol reaches a graduated scale.
- Temperature is indicated by the changes in liquid.
- Alcohol In Glass Thermometer features coloured alcohol due to the dangerous contact with mercury.

Constant Pressure Gas Thermometer

- A constant pressure gas thermometer can function at constant pressure and volume conditions.
- Constant pressure gas thermometer offers highly accurate measurement and is used to adjust other thermometers.

Constant Volume Gas Thermometer

- In a constant volume gas thermometer, the pressure increases when the temperature increases.
- Constant volume gas thermometer measures temperature in the range 0k to 500 k.

Platinum Resistance Thermometer

- Platinum Resistance Thermometers use a platinum wire attached to an electrical resistance to measure the temperature.
- Generally, these thermometers are used to measure outside temperatures.
- Platinum Resistance Thermometers are slow and offer accurate readings.
- The material's resistance in the thermometer increases when the temperature increases.
- Platinum resistance thermometers measure temperature in the range of 500 k to 2300 k.

The resistance of the material in the thermometer increases when the temperature increases. Platinum resistance thermometers measure temperature in the range of 500 k to 2300 k.

Thermocouple Thermometer

- These devices are used for the quick measurement of temperature.
- It finds its application in laboratories.
- Electrical resistance is used in the device to generate a voltage that varies according to the connection temperature.
- It also measures temperature in the range of 500 k to 2300 k.

Pyrometer Thermometer

- The thermometer operates on Stefan's law.

- This device measures the temperature from the heat radiation emitted by the objects.
- These thermometers can be used without being in contact with the objects.
- A pyrometer Thermometer measures a temperature greater than 2000 K.

Temperature conversion exercise

1. $45^{\circ}\text{R} = \dots^{\circ}\text{F} = \dots^{\circ}\text{C} = \dots\text{K}$
2. $70^{\circ}\text{C} = \dots^{\circ}\text{F} = \dots^{\circ}\text{R} = \dots\text{K}$
3. $293\text{K} = \dots^{\circ}\text{F} = \dots^{\circ}\text{C} = \dots^{\circ}\text{R}$
4. $77^{\circ}\text{F} = \dots^{\circ}\text{R} = \dots^{\circ}\text{C} = \dots\text{K}$
5. The temperature of ice that is melting and the temperature of boiling water when measured with an Agus thermometer are 10°A and 160°A respectively. The temperature of an object is measured with a Celsius scale thermometer of 50°C . What is the temperature of the object if measured with an Agus thermometer?
6. Susi made a thermometer, it turns out that according to her the temperature when ice melts is -20° and the temperature when water boils is 180° . If Susi's thermometer shows a temperature of 60° , what temperature is the Fahrenheit thermometer showing at that time?
7. Bernard makes a thermometer, it turns out that the temperature at which ice melts is -15° and the temperature at which water boils is 150° . If Bernard's thermometer shows a temperature of 75° , what temperature is the Kelvin thermometer showing at that time?

Worksheet

Temperature

1. Aim Practical

- To measure the temperature of objects using a thermometer
- To convert temperature to Reamur, Fahrenheit and Kelvin scales

2. Basic theory

Changing the temperature scale can be done by using the conversion formula as follows

$$\text{TC} : \text{TR} : \text{TF} - 32 : \text{Tk} - 273 = 5 : 4 : 9 : 5$$

3. Tools and materials

Tools =

Materials =

4. Procedures

1. Prepare ice cubes and water then put it in the beaker glass.
2. Take a thermometer and insert it into a glass containing ice cubes and water
3. Measure the temperature of the ice cubes and water using a thermometer

No	Suhu benda	Celcius	Reamur	Fahrenheit	Kelvin
1	Ice				
2	Water				

Chapter 3: Heat

In this chapter you will:

- to describe the definition of heat
- to observe factors that can speed up evaporation
- to observe the amount of heat needed to raise the temperature of a substance
- to define three methods of heat transfer



Have you ever made a cup of tea? When hot water is poured into the cup, the cup becomes warm or even hot. In this condition, the transfer of heat has occurred. Does this mean that the terms heat and hot temperature are the same?

A. The definition of Heat

What is heat? In order to get a better understanding of the definition of heat, do the following experiment.

The Definition of Heat

Objective: To understand the definition of heat

Apparatus and Materials: 1. A metal spoon

2. a glass

3. hot water

Procedure:

Touch the glass before it is filled with hot water.

Pour hot water into the glass carefully.

Touch the spoon before it is put into the glass.

Put the spoon into the glass filled with hot water.

After a moment, touch the glass and the spoon, What do you feel?

Make conclusions from your activity.

In the experiment, we find that the spoon becomes warm after being put into a glass of hot water. The glass also becomes warm or its temperature rises after being filled with hot water.

This phenomenon proves that there is energy transfer from material with a higher temperature to material with a lower temperature. This transferred energy is called *heat*. Thus it can be concluded that:

Heat is the state of energy which is transferred from a higher temperature to a lower temperature.

Heat is quantity, therefore heat has a unit. The unit of heat is a calorie. In the International System of units, The unit of heat is a joule (J).

1 joule = 0.24 calories

Another example proving that heat is energy which is transferred from a higher temperature to a lower temperature, is when you drink cold or warm water.

When you drink cold water, your tongue feels cold because the tongue has a higher temperature than the cold water, so the heat energy from the tongue is transferred to the cold water. When you drink hot tea, your tongue feels hot because hot tea has a higher temperature than that of the tongue. Therefore, the heat energy is transferred to your tongue.

B. How heat can Raise the temperature of a Substance and Change the State of Matter

Because of heat, material may go through a temperature rise or go through a change of state of matter. In order to observe it, pay attention to the following discussion.

1. Heat can raise the temperature of a substance

To find out that heat can raise the temperature of a substance, try the following experiment.

Heat can Raise the Temperature of a Substance

Objective: To observe that heat can raise the temperature of substance

Apparatus and Materials:

- Beaker glass
- A tripod, gauze and bunsen burner
- A thermometer
- Water
- Ice cubes

Procedure:

Arrange the apparatus as shown in the picture.

Fill the Beaker glass with ice cubes and water, then measure the temperature with your thermometer.

Put the glass on tripod, then light the Bunsen burner.

Heat the ice cubes and water until they are boiling, note the temperature change.

Make the conclusions from your experiment.

In the experiment, the ice which is heated will turn into water (melting). When it melts, the temperature is 0°C . After the ice turns into water, the water temperature will keep on rising until it is boiling. In this boiling phenomenon, the water transforms into steam. When the water boils, the thermometer scale will stay at the point of 100°C , although the water continues to boil.

The transformation that occurs to the ice indicates that if heat is applied to an object then the object will go through a change in temperature or a change in state of matter. *When the state of matter changes, the temperature does not change and when the temperature changes, the state of matter does not change.*

Based on the measurement, the heat needed by 1 kg of water to achieve a 1°C temperature rise is 4,200 joules, meanwhile the heat needed by 1 kg of alcohol to achieve a 1°C temperature rise is 2,400 joules. The amount of heat needed by 1 kg of substance for every 1°C of temperature rise is called *specific heat*. So, *specific heat is the amount of heat needed by one mass of substance to raise the temperature by 1°C*. Meanwhile the specific heat of alcohol is 2,400 J/kg°C. Thus it can be noted that:

The amount of heat needed is proportional to the temperature rise, the mass of substance and the specific heat of substance.

The correlation between heat needed, temperature rise, the mass of a substance and the specific heat of substance can be formulated as:

The amount of heat needed to raise the temperature = mass of substance x specific heat of substance x temperature rise. Or,

$$Q = m \times c \times \Delta T$$

where:

Q = amount of heat, unit: joule (J)

m = mass, unit: kg

c = specific heat, unit: J/kg°C

ΔT = temperature change, unit: °C

The specific heat of some substances are as follows:

Substances	Specific Heat (J/kg°C)	Substance	Specific Heat (J/kg°C)
Water	4,200	Marble	860
Ice	2,100	Glass	670
Human body	3,470	Iron or Steel	450
Alcohol	2,400	Copper	390
Wood	1,700	Silver	230
Air	1,000	Lead	130
Aluminium	900		

Example:

- How much heat is needed if 2 kg of water is heated from 10°C to 60°C? (the specific heat of water is 4,200 J/kg°C)

Solution:

m = 2 kg

$\Delta T = 60^\circ\text{C} - 10^\circ\text{C} = 50^\circ\text{C}$

c = 4,200 J/kg°C

Q....?

Answer:

$Q = m \times c \times \Delta T$

= 2 kg x 4,200 J/kg°C x 50°C

= 420,000 joule

= 420 kJ

Therefore, the heat needed is 420 kJ.

2. Heat changes the state of matter

When heat is added to a substance, the molecules gain energy and vibrate and force other molecules apart. As a result expansion takes place. You would have seen some space being left in railway tracks. It is because, during summer time, more heat causes expansion in tracks. Expansion is greater for liquids than for solids and maximum in case of gases.



a. Change in temperature

When heat energy is added to a substance, the kinetic energy of its particles increases and so the particles move at higher speed. This causes rise in temperature. When a substance is cooled, that is, when heat is removed, the molecules lose heat and its temperature falls.

b. Change in state

When you heat ice cubes, they become water and water on further heating changes into vapour. So, solid becomes liquid and liquid becomes gas, when heat is added. The reverse takes place when heat is removed.

c. Chemical changes

Since heat is a form of energy it plays a major role in chemical changes. In some cases, chemical reactions need heat to begin and also heat determines the speed at which reactions occur. When we cook food, we light the wood and it catches fire and the food particles become soft because of the heat energy. These are all the chemical changes taking place due to heat.

The discussion above and the accompanying equation ($Q = m \cdot C \cdot \Delta T$) relates the heat gained or lost by an object to the resulting temperature changes of that object. As we have learned, sometimes heat is gained or lost but there is no temperature change. This is the case when the substance is undergoing a state change. So now we must investigate the mathematics related to changes in state and the quantity of heat.

To begin the discussion, let's consider the various state changes that could be observed for a sample of matter. The table below lists several state changes and identifies the name commonly associated with each process.

Process	Change of State
Melting	Solid to Liquid
Freezing	Liquid to Solid
Vaporization	Liquid to Gas
Condensation	Gas to Liquid
Sublimation	Solid to Gas
Deposition	Gas to Solid

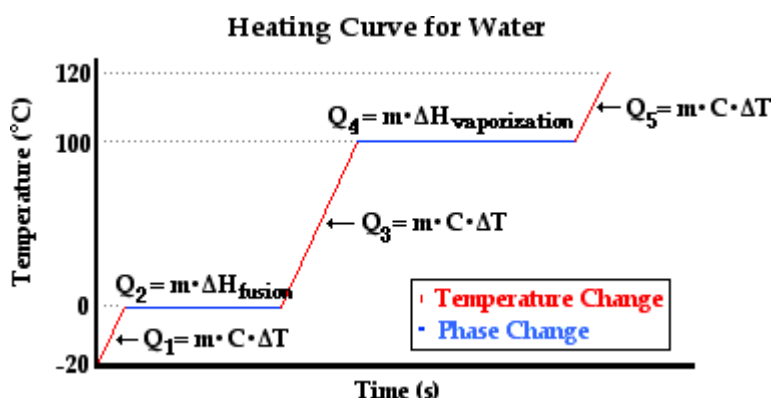
In the case of melting, boiling and sublimation, energy would have to be added to the sample of matter in order to cause the change of state. Such state changes are referred to as being **endothermic**. Freezing, condensation

and deposition are **exothermic**; energy is released by the sample of matter when these state changes occur. So one might notice that a sample of ice (solid water) undergoes melting when it is placed on or near a burner. Heat is transferred from the burner to the sample of ice; energy is gained by the ice causing the change of state. But how much energy would be required to cause such a change of state? Is there a mathematical formula that might help in determining the answer to this question? There most certainly is.

The amount of energy required to change the state of a sample of matter depends on three things. It depends upon what the substance is, on how much substance is undergoing the state change, and upon what state change that is occurring. For instance, it requires a different amount of energy to melt ice (solid water) compared to melting iron. And it requires a different amount of energy to melt ice (solid water) as it does to vaporize the same amount of liquid water. And finally, it requires a different amount of energy to melt 10.0 grams of ice compared to melting 100.0 grams of ice. The substance, the process and the amount of substance are the three variables that affect the amount of energy required to cause a specific change in state. Use the widget below to investigate the effect of the substance and the process upon the energy change. (Note that the Heat of Fusion is the energy change associated with the solid-liquid state change.)

On the [previous page of Lesson 2](#), the heating curve of water was discussed. The heating curve showed how the temperature of water increased over the course of time as a sample of water in its solid state (i.e., ice) was heated. We learned that the addition of heat to the sample of water could cause either changes in temperature or changes in state. At the melting point of water, the addition of heat causes a transformation of the water from the solid state to the liquid state. And at the boiling point of water, the addition of heat causes a transformation of the water from the liquid state to the gaseous state. These changes in state occurred without any changes in temperature. However, the addition of heat to a sample of water that is not at any phase change temperatures will result in a change in temperature.

Now we can approach the topic of heating curves on a more quantitative basis. The diagram below represents the heating curve of water. There are five labeled sections on the plotted lines.



The three diagonal sections represent the changes in temperature of the sample of water in the solid state (section 1), the liquid state (section 3), and the gaseous state (section 5). The two horizontal sections represent the changes in state of the water. In section 2, the sample of water is undergoing melting; the solid is changing to a liquid. In section 4, the sample of water is undergoing boiling; the liquid is changing to a gas. The quantity of heat transferred to the water in sections 1, 3, and 5 is related to the mass of the sample and the temperature change by the formula $Q = m \cdot C \cdot \Delta T$. And the quantity of heat transferred to the water in sections 2 and 4 is related to the mass of the sample and the heat of fusion and vaporization by the formulae $Q = m \cdot \Delta H_{\text{fusion}}$ (section 2) and $Q = m \cdot \Delta H_{\text{vaporization}}$ (section 4). So now we will make an effort to calculate the quantity of heat required to change 50.0 grams of water from the solid state at -20.0°C to the gaseous state at 120.0°C . The calculation will require five steps - one step for each section of the above graph. While the specific heat capacity of a substance varies with temperature, we will use the following values of specific heat in our calculations:

Solid Water: $C = 2.00 \text{ J/g}^{\circ}\text{C}$

Liquid Water: $C = 4.18 \text{ J/g}^{\circ}\text{C}$

Gaseous Water: $C = 2.01 \text{ J/g}^{\circ}\text{C}$

Finally, we will use the previously reported values of ΔH_{fusion} (333 J/g) and $\Delta H_{\text{vaporization}}$ (2.23 kJ/g).

Section 1: Changing the temperature of solid water (ice) from -20.0°C to 0.0°C .

Use $Q_1 = m \cdot C \cdot \Delta T$

where $m = 50.0\text{ g}$, $C = 2.00\text{ J/g/}^{\circ}\text{C}$, $T_{\text{initial}} = -20.0^{\circ}\text{C}$, and $T_{\text{final}} = 0.0^{\circ}\text{C}$

$$Q_1 = m \cdot C \cdot \Delta T = (50.0\text{ g}) \cdot (2.00\text{ J/g/}^{\circ}\text{C}) \cdot (0.0^{\circ}\text{C} - -20.0^{\circ}\text{C})$$

$$Q_1 = 2.00 \times 10^3\text{ J} = 2.00\text{ kJ}$$

Section 2: Melting the Ice at 0.0°C .

Use $Q_2 = m \cdot \Delta H_{\text{fusion}}$

where $m = 50.0\text{ g}$ and $\Delta H_{\text{fusion}} = 333\text{ J/g}$

$$Q_2 = m \cdot \Delta H_{\text{fusion}} = (50.0\text{ g}) \cdot (333\text{ J/g})$$

$$Q_2 = 1.665 \times 10^4\text{ J} = 16.65\text{ kJ}$$

$$Q_2 = 16.7\text{ kJ (rounded to 3 significant digits)}$$

Section 3: Changing the temperature of liquid water from 0.0°C to 100.0°C .

Use $Q_3 = m \cdot C \cdot \Delta T$

where $m = 50.0\text{ g}$, $C = 4.18\text{ J/g/}^{\circ}\text{C}$, $T_{\text{initial}} = 0.0^{\circ}\text{C}$, and $T_{\text{final}} = 100.0^{\circ}\text{C}$

$$Q_3 = m \cdot C \cdot \Delta T = (50.0\text{ g}) \cdot (4.18\text{ J/g/}^{\circ}\text{C}) \cdot (100.0^{\circ}\text{C} - 0.0^{\circ}\text{C})$$

$$Q_3 = 2.09 \times 10^4\text{ J} = 20.9\text{ kJ}$$

Section 4: Boiling the Water at 100.0°C .

Use $Q_4 = m \cdot \Delta H_{\text{vaporization}}$

where $m = 50.0\text{ g}$ and $\Delta H_{\text{vaporization}} = 2.23\text{ kJ/g}$

$$Q_4 = m \cdot \Delta H_{\text{vaporization}} = (50.0\text{ g}) \cdot (2.23\text{ kJ/g})$$

$$Q_4 = 111.5\text{ kJ}$$

$$Q_4 = 112\text{ kJ (rounded to 3 significant digits)}$$

Section 5: Changing the temperature of liquid water from 100.0°C to 120.0°C .

Use $Q_5 = m \cdot C \cdot \Delta T$

where $m = 50.0\text{ g}$, $C = 2.01\text{ J/g/}^{\circ}\text{C}$, $T_{\text{initial}} = 100.0^{\circ}\text{C}$, and $T_{\text{final}} = 120.0^{\circ}\text{C}$

$$Q_5 = m \cdot C \cdot \Delta T = (50.0\text{ g}) \cdot (2.01\text{ J/g/}^{\circ}\text{C}) \cdot (120.0^{\circ}\text{C} - 100.0^{\circ}\text{C})$$

$$Q_5 = 2.01 \times 10^3\text{ J} = 2.01\text{ kJ}$$

The total amount of heat required to change solid water (ice) at -20°C to gaseous water at 120°C is the sum of the Q values for each section of the graph. That is,

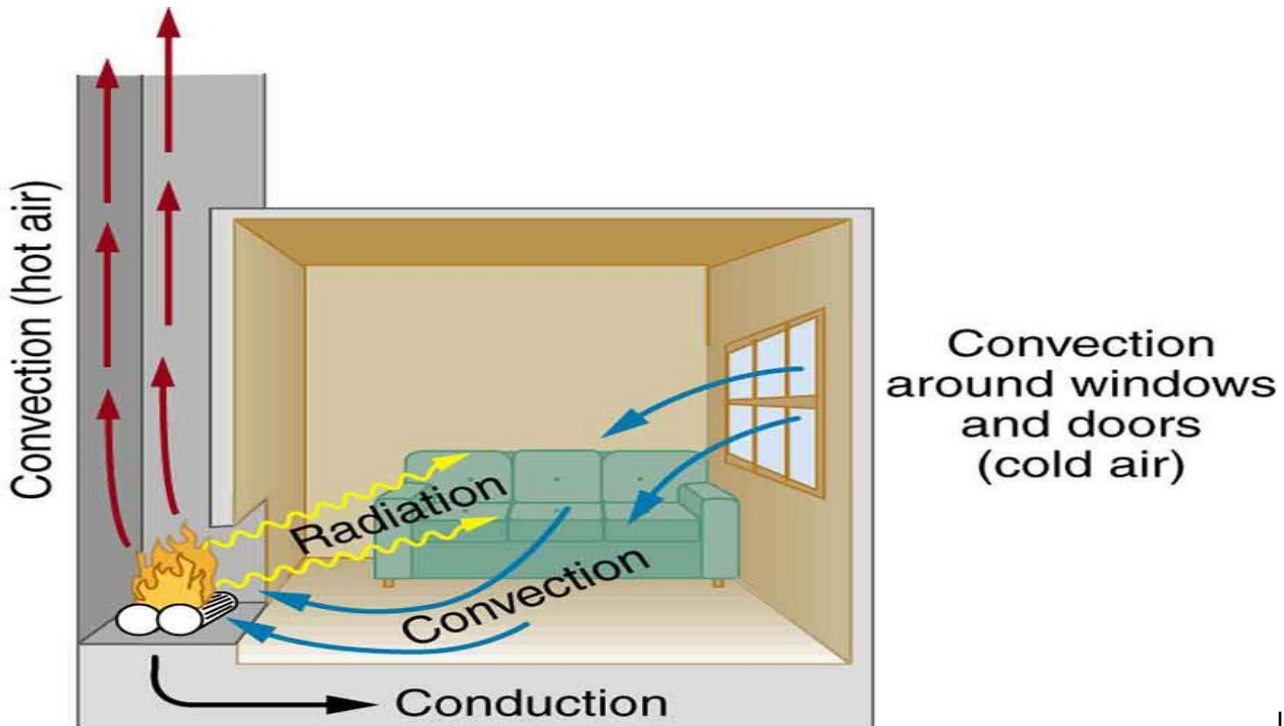
$$Q_{\text{total}} = Q_1 + Q_2 + Q_3 + Q_4 + Q_5$$

Summing these five Q values and rounding to the proper number of significant digits leads to a value of **154 kJ** as the answer to the original question.

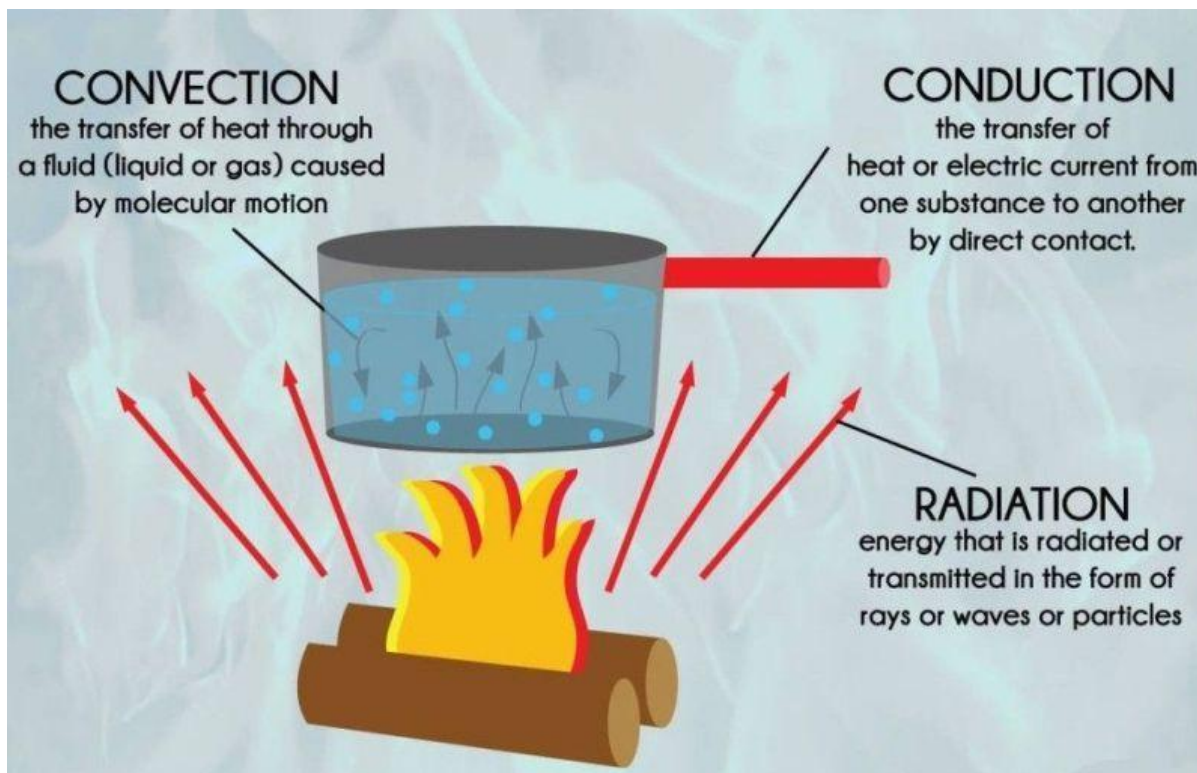
C. Three Methods of Heat Transfer

Equally as interesting as the effects of heat transfer on a system are the methods by which this occurs. Whenever there is a temperature difference, heat transfer occurs. Heat transfer may occur rapidly, such as through a cooking pan, or slowly, such as through the walls of a picnic ice chest. We can control rates of heat transfer by choosing materials (such as thick wool clothing for the winter), controlling air movement (such as the use of weather stripping around doors), or by choice of color (such as a white roof to reflect summer sunlight). So many processes involve heat transfer, so that it is hard to imagine a situation where no heat transfer occurs. Yet every process involving heat transfer takes place by only three methods:

1. **Conduction** is heat transfer through stationary matter by physical contact. (The matter is stationary on a macroscopic scale—we know there is thermal motion of the atoms and molecules at any temperature above absolute zero.) Heat transferred between the electric burner of a stove and the bottom of a pan is transferred by conduction.
2. **Convection** is the heat transfer by the macroscopic movement of a fluid. This type of transfer takes place in a forced-air furnace and in weather systems, for example.
3. Heat transfer by **radiation** occurs when microwaves, infrared radiation, visible light, or another form of electromagnetic radiation is emitted or absorbed. An obvious example is the warming of the Earth by the Sun. A less obvious example is thermal radiation from the human body.



In a fireplace, heat transfer occurs by all three methods: conduction, convection, and radiation. Radiation is responsible for most of the heat transferred into the room. Heat transfer also occurs through conduction into the room, but at a much slower rate. Heat transfer by convection also occurs through cold air entering the room around windows and hot air leaving the room by rising up the chimney.



D. Applications of Heat in Daily Life

- Drinks can be cooled by adding in several cubes of ice. When ice melts a large amount of heat is absorbed and this lowers the temperature of the drink.



- The freshness of fish and meat can be maintained by placing them in contact with ice. With its larger latent heat, ice is able to absorb a large quantity of heat from the fish as it melts. Thus, food can be kept at a low temperature for an extended period of time.



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- Water has a large specific latent heat of vaporization. This property enables steam to be used for cooking by the method of steaming. When steam condenses on the food, the latent heat is released directly onto the food enables the food to be cooked at a faster rate.



- Our bodies feel cool after sweating. This is because latent heat of vaporization is absorbed from the body when sweat evaporates. As a result, the body is cooled by the removal of heat.

- Always be very careful when opening the lid of a pot when the water in it is boiling. Water has a large specific latent heat of vaporization. When steam condenses on the skin of your arm, the very large amount of latent heat released can cause a serious burn.



The heating effect of electric current is used in electrical appliances like electric heater, electric iron, electric room heater, immersion heater electric kettle, hair dryer etc.

All these appliances have a coil of wire called an element. When electric current flows through the element it becomes hot and gives out heat. The amount of heat produced in a wire depends upon its material, length and thickness.

Electric Heater



Electric Iron



Electric Room Heater



Exercise

1. 3 kg of silver is heated until it becomes molten silver at 961°C . If the heat of melting silver is $88,000 \text{ J/kg}$, how much heat is required?
2. A 500 gram mass of mercury is heated to a gaseous state at 357°C . If the heat of mercury vapor is $272,000 \text{ J/kg}$, how much heat is needed by mercury?
3. Marble with a mass of 20 kg is heated from 20°C to 80°C . If the specific heat of marble is $860 \text{ J/kg } ^{\circ}\text{C}$, how much heat is needed for the marble?
4. A container containing 5 kg of water has an initial water temperature of 10°C and is heated to a water temperature of 100°C . If the specific heat of water is $4200 \text{ J/kg } ^{\circ}\text{C}$, how much heat is needed to heat the water?
5. Heat can move from high temperature to low temperature. An object that conducts heat well is called and give 2 examples of an object that conducts heat well
6. When Chakra is drying clothes in a hot place, the clothes will dry because of the hot sun. In this event, what heat transfer occurs and give 2 other examples in everyday life?
7. Dhira did the practice of heating water in the physics laboratory. What heat transfer occurs during this practice and provide an explanation?

Heat practice worksheet

1. Aim of the Experiment

Through discussions and experiments, students can explain the effect of heat on temperature and the state of objects and are skilled at conducting experiments investigating the effects of heat on temperature and state of objects.

2. Apparatus

- a. ice
- b. bunsen
- c. 3 leg support
- d. timer
- e. beaker glass
- f. balance

3. Procedures / Ways of Working

1. Fill a beaker with 100 grams of ice on a balance
2. Place a beaker filled with water on the footrest 3 insert it thermometer into the beaker.
3. Turn on the bunsen and insert the bunsen under the leg support on top of it is a beaker filled with ice. observe temperature changes every minute for up to 15 minutes.
4. Record the water temperature every minute on the experimental results table and make a graph with time on the x-axis and temperature on the y-axis.

4. Result of Practice

Time (minute)	Suhu (°C)	Time (minute)	Suhu (°C)
0		6	
1		7	
2		8	
3		9	
4		10	
5			

5. Make a graph of time in minutes on the x-axis and temperature in °C on the y-axis. Make a graph of time in minutes on the x-axis and temperature

6. Conclusion

1. Ice completely melts at
2. Water vapor is visible at

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